Reviewer Pack — Minimal Geometric Entropy of Riemann Surfaces and SOS Max-CU…

Entry 4a4f69a6f8ba · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Geometric Entropy of Riemann Surfaces and SOS Max-CUT Approximation

Entry ID: `4a4f69a6f8ba` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-21 19:37:02 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry: Riemann Surfaces **Field B** (complexity object): Sum-of-squares (SOS) Hierarchy for Max-CUT

Statement:

> ['Let M be a degree-d pseudoexpectation matrix of an n-vertex graph. The minimal geometric entropy of the associated Riemann surface is bounded by O(n^(1/d/2)) if and only if there exists a d-degree SOS certificate that approximates max-CUT within a factor of 0.878.', 'For all n ≤ 40, any pseudoexpectation matrix M satisfying this condition must have a corresponding SOS certificate with at most d variables.']

Rationale (proposer's reasoning):

> ["Riemann surfaces provide a geometric interpretation of the eigenvalues of the adjacency matrix, which can be used to define an entropy measure. This entropy might capture information about the graph's structure that is relevant to max-CUT approximation.", 'The SOS hierarchy has been shown to be related to max-CUT through the Goemans-Williamson algorithm, suggesting a potential connection between geometric entropies and SOS approximations.']

Taxonomy category: `SOS_DEGREE` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `573594a69bdbbe81`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if for all n ≤ 40, the geometric entropy of the Riemann surface associated with a pseudoexpectation matrix M is O(n^(1/d/2)) AND an SOS certificate constructed from M approximates max-CUT within a factor of 0.878 using the same number of variables as M.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Riemann surfaces" AND "Sum-of-squares (SOS) Hierarchy for Max-CUT" - "minimal geometric entropy" AND "pseudoexpectation matrix" - "SOS max-CUT approximation" AND "Riemann surface"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math
from fractions import Fraction
from itertools import combinations, permutations

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_graph(n):
        edges = set()
        for u in range(n):
            for v in range(u + 1, n):
                if random.random() < 0.5:
                    edges.add((u, v))
        return {u: [v for v in range(n) if (u, v) in edges or (v, u) in edges] for u in range(n)}

    def eigenvalues(matrix):
        n = len(matrix)
        A = matrix.copy()
        for i in range(n):
            pivot_row = max(range(i, n), key=lambda r: abs(A[r][i]))
            if A[pivot_row][i] == 0:
                continue
            A[pivot_row], A[i] = A[i], A[pivot_row]
            for j in range(n):
                if j != i:
                    factor = -A[j][i] / A[i][i]
                    A[j] = [x + factor * y for x, y in zip(A[j], A[i])]
        eigenvals = []
        for row in A:
```

```
            if all(x == 0 for x in row[:n-1]):
                eigenvals.append(row[-1])
        return eigenvals

    def geometric_entropy(eigenvals):
        return sum(-x * math.log2(x) for x in eigenvals if x > 0)

    def sos_certificate(matrix, d):
        n = len(matrix)
        variables = set()
        for u in range(n):
            for v in range(u + 1, n):
                if matrix[u][v] != 0:
                    variables.add((u, v))
        return variables

    def max_cut_approximation(graph, certificate):
        cut_value = sum(1 for (u, v) in graph if (u, v) in certificate)
        return cut_value / len(graph)

    n = random.randint(5, 40)
    d = random.randint(2, 3)
    M = generate_graph(n)
    eigenvals = eigenvalues(M)
    entropy = geometric_entropy(eigenvals)
    sos_cert = sos_certificate(M, d)
    max_cut_approx = max_cut_approximation(M, sos_cert)

    return {
        "metric_name": "geometric_entropy",
        "metric_value": entropy,
        "instances_tested": 1,
        "conjecture_holds": entropy <= n ** (1 / (d / 2)) and max_cut_approx >= 0.878 * len(sos_ce
        "counterexample": "" if entropy <= n ** (1 / (d / 2)) and max_cut_approx >= 0.878 * len(sc
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] if sys.argv[1:] else [2, 3, 5, 7, 11, 13, 17, 19, 23, 2
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_entropy = sum(r["metric_value"] for r in results) / len(results)
    std_entropy = math.sqrt(sum((r["metric_value"] - mean_entropy) ** 2 for r in results) / len(re
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_entropy} std={std_entropy} support_fraction={support_
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_entropy} std={std_entropy} support_fraction={support_
    else:
        for r in results:
```

```
            if not r["conjecture_holds"]:
                print(f"RESULT: FALSIFIED counterexample=\"{r['counterexample']}\" first_failing_s
                break
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5bacaf8.py", line 84, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5bacaf8.py", line 65, in run_trial
    eigenvals = eigenvalues(M)
                ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5bacaf8.py", line 32, in eigenvalues
    pivot_row = max(range(i, n), key=lambda r: abs(A[r][i]))
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5bacaf8.py", line 32, in <lambda>
    pivot_row = max(range(i, n), key=lambda r: abs(A[r][i]))
                                                   ~~~~^^^
IndexError: list index out of range
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test code crashed before producing data, which prevents us from verifying whether the conjecture's conditions are met. | next: Investigate and fix the crash in the test code to verify the conjecture's conditions.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14962
2	preregistration	ollama_remote	glm4:latest	0	0	9570
3	novelty	ollama_remote	glm4:latest	0	0	8358
4	novelty	ollama_remote	glm4:latest	0	0	9603
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14706
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13361
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18961

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11093
9	judge	ollama_remote	glm4:latest	0	0	12437

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 113050 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/4a4f69a6f8ba.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/4a4f69a6f8ba.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/4a4f69a6f8ba.tar.gz` (if generated)